SOUTHLAKE AGENTIC SYNTHETIC DATA STUDIO

From Public ED Data to Southlake Future-State Planning Scenarios

An agentic workflow — with reasoning and analysis technically supported by **theFinlyApp** — that turns safe public emergency department data into synthetic planning artifacts, then makes the limitations visible.

WHAT JUDGES SHOULD KNOW

The emphasis is on **how the system thinks and acts**, not just the data it produces. The agentic reasoning engine is technically supported by theFinlyApp — powering every step from intent framing to quality evaluation and self-correction.

1 Intent Agent

Interprets the stakeholder's planning goal and frames it as a specific operational question.

AI REASONING

→

2 Profile Agent

Analyzes dataset quality, identifies sensitive columns, flags missingness.

AI REASONING

→

3 Strategy Agent

Decides synthesis approach, target rows, and constraints — not scripted.

AI DECIDES

→

4 Evaluate Agent

Scores fidelity, privacy, utility. Triggers retry if quality is below threshold.

AI SELF-CORRECTS

→

5 Narrative Agent

Crafts planning summaries tailored to the scenario and audience.

AI NARRATES

INPUT

CDC/NCHS NHAMCS 2022 public ED dataset — 13,085 visits, 21 operational columns

SOUTHLAKE SCENARIOS

Distributed campus routing · Community diversion · Regional growth · ED surge

OUTPUT

Synthetic dataset + evaluation metrics + caution report + pitch artifacts + downloadable ZIP

WHAT MAKES IT AGENTIC

AI reasons and decides at each step · Strategy adapts to data · Evaluate triggers retries · Powered by theFinlyApp

WHAT THE STUDIO DOES

A Planning Studio, Not Just a Data Generator

Saved demo reliability, visible AI reasoning, Southlake-specific scenarios, and exportable planning artifacts.

Southlake Scenarios

Four planning scenarios aligned to distributed-network strategy: campus routing, diversion, growth, surge.

Visible AI Reasoning

Reasoning trace at every agent step — judges see what the AI decided and why.

Evaluation Dashboard

Fidelity, privacy, utility scores plus source-vs-synthetic comparison in one screen.

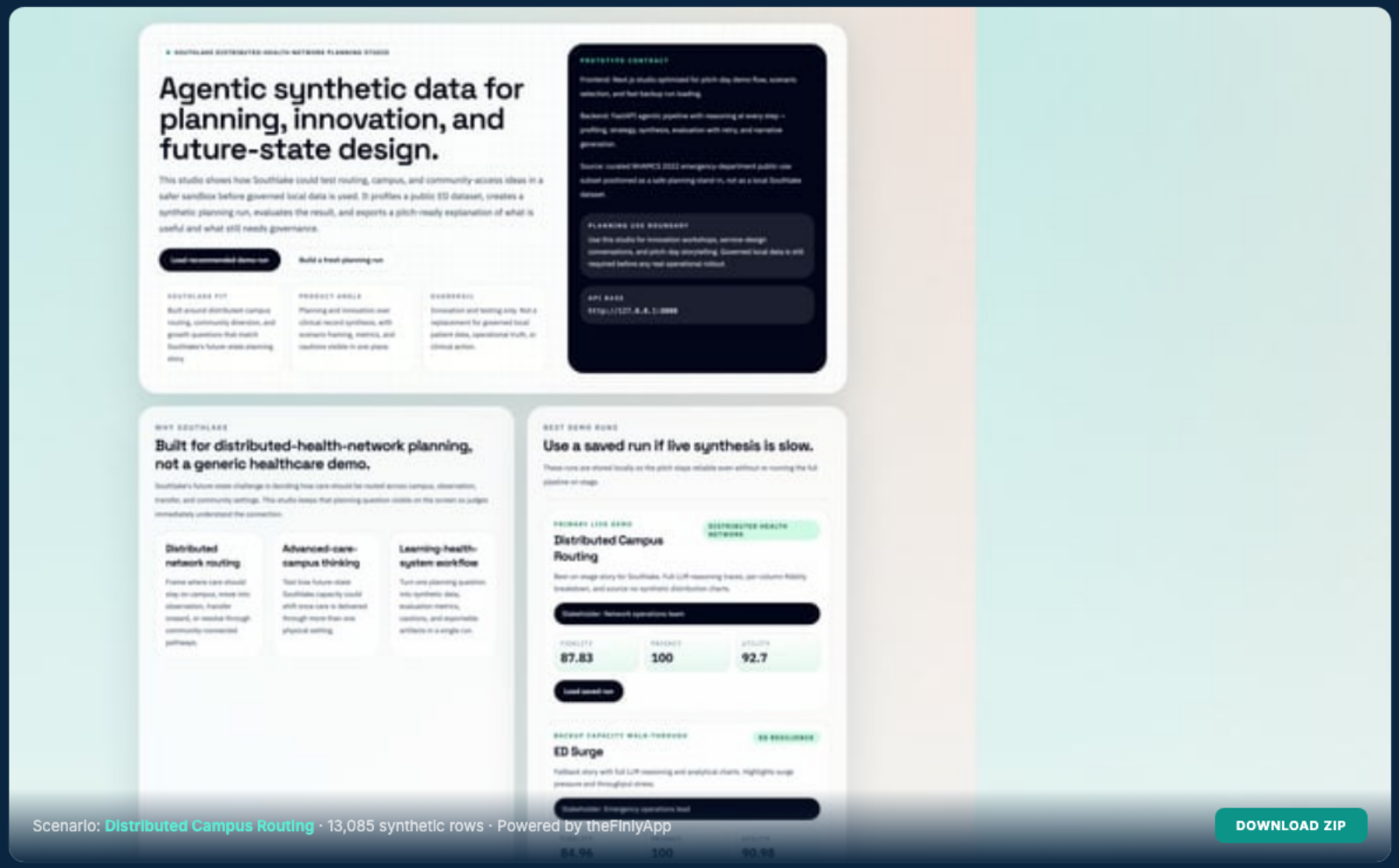
Export & Narrative

Auto-generated methodology, features, cautions + downloadable ZIP for workshops.

Reasoning and analysis technically supported by **theFinlyApp** — a financial intelligence platform whose agentic architecture was adapted for healthcare planning.

LIVE DEMO OUTPUT — DISTRIBUTED CAMPUS ROUTING

FIDELITY	PRIVACY	UTILITY	EXACT OVERLAP
87.8	100	92.7	0%



Scenario: Distributed Campus Routing · 13,085 synthetic rows · Powered by theFinlyApp

DOWNLOAD ZIP

PLANNING USE BOUNDARY

What the Studio Is Safe For, and What It Is Not

Synthetic data is powerful for planning and experimentation. But it is not local operational truth, and it should not be used without governance and validation.

Safe For

- Planning workshops and innovation sprints — test routing, capacity, and community-diversion ideas safely
- Scenario-driven discussions about Southlake's distributed-campus future state
- Prototyping dashboards, workflows, and analytics without real patient records
- Pitch-day storytelling backed by visible metrics, AI reasoning, and exportable artifacts

Not Ready For

- Clinical care, staffing commitments, reimbursement, or patient-level action
- Claims about Southlake's real patient mix from a U.S. public-use source
- A full hospital digital twin or validated multi-table operational model
- Production use without governance, privacy review, and validation on governed local data

BEST NEXT STEP

Validate this workflow with governed Southlake data, operational owners, and formal privacy review. The agentic architecture is ready — it needs real local data to become a real planning tool.

WHY THIS SLIDE MATTERS

In healthcare, the team that says "here's what we don't know yet" is more trustworthy than the team that claims everything is solved. This caution boundary is a feature, not a weakness.